

2.1.3 Amount of substance

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
The mole	
(a) explanation and use of the terms: (i) <i>amount of substance</i> (ii) <i>mole</i> (symbol 'mol'), as the unit for amount of substance (iii) the <i>Avogadro constant</i> , N_A (the number of particles per mole, $6.02 \times 10^{23} \text{ mol}^{-1}$) (iv) <i>molar mass</i> (mass per mole, units g mol^{-1}), (v) <i>molar gas volume</i> (gas volume per mole, units $\text{dm}^3 \text{ mol}^{-1}$)	<i>M0.0, M0.1, M0.2, M0.4</i> Amount of substance will be used in exams using the formula of the substance, e.g. amount of NaCl; amount of O_2 . In recognition of IUPAC's review, we will accept both the classical (carbon-12 based) and revised (Avogadro constant based) definitions of the mole in examinations from June 2018 onwards (see https://iupac.org/new-definition-mole-arrived/) The value for N_A and the molar gas volume at RTP are provided on the <i>Data Sheet</i> .
Determination of formulae	
(b) use of the terms: (i) <i>empirical formula</i> (the simplest whole number ratio of atoms of each element present in a compound) (ii) <i>molecular formula</i> (the number and type of atoms of each element in a molecule)	Definitions not required.
(c) calculations of empirical and molecular formulae, from composition by mass or percentage compositions by mass and relative molecular mass	<i>M0.2, M2.2, M2.3, M2.4</i> To include calculating empirical formulae from elemental analysis data (see also 6.3.2 e).
(d) the terms <i>anhydrous</i> , <i>hydrated</i> and <i>water of crystallisation</i> and calculation of the formula of a hydrated salt from given percentage composition, mass composition or based on experimental results	<i>M0.2, M2.2, M2.3, M2.4</i> PAG1
Calculation of reacting masses, gas volumes and mole concentrations	
(e) calculations, using amount of substance in mol, involving: (i) mass (ii) gas volume (iii) solution volume and concentration	<i>M0.0, M0.1, M0.4, M1.1, M2.2, M2.3, M2.4</i> Learners will be expected to express concentration in mol dm^{-3} and g dm^{-3} .
(f) the ideal gas equation: $pV = nRT$	<i>M0.0, M0.1, M0.4, M1.1, M2.2, M2.3, M2.4</i> The value for R is provided on the <i>Data Sheet</i> . Learners will be expected to express quantities in SI units.
(g) use of stoichiometric relationships in calculations	<i>M0.2</i>

Percentage yields and atom economy

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| <p>(h) calculations to determine:</p> <ul style="list-style-type: none">(i) the percentage yield of a reaction or related quantities(ii) the atom economy of a reaction <p>(i) the techniques and procedures required during experiments requiring the measurement of mass, volumes of solutions and gas volumes</p> <p>(j) the benefits for sustainability of developing chemical processes with a high atom economy.</p> | <p><i>M0.2, M1.1, M2.2, M2.3, M2.4</i></p>
<p>PAG1
HSW4 Many opportunities to carry out experimental and investigative work.</p>
<p>HSW10 Use of processes with high atom economy in chemical industry and other areas.</p> |
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